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Project Report

DriveSafe AI: Automated FMCSA Compliance Auditor

Abstract and Introduction

This project focuses on building a Natural Language Processing (NLP) model to detect compliance violations in trucking log entries. The main goal is to create an automated system that can read text and classify it into different categories such as safe logs and different types of violations.

In the trucking industry, companies must follow strict rules set by the Federal Motor Carrier Safety Administration (FMCSA). Drivers are required to follow Hours of Service (HOS) regulations and maintain accurate records. Checking these logs manually takes a lot of time and can lead to human errors. Because of this, there is a need for an automated solution that can help improve accuracy and efficiency.

In this project, a Transformer-based model called DistilBERT is used to understand and classify text data. The model is trained on a dataset that includes different types of log entries, such as SAFE, HOS_VIOLATION, FORM_ERROR, MISSING_DATA, and INSPECTION_ISSUE. After training, the model can predict the correct category for new log entries.

The results show that the model performs well, achieving an accuracy of 94% and a weighted F1-score of 0.94. This means the model can correctly classify most of the log entries and works effectively for this task. The system can help trucking companies reduce manual work, detect violations faster, and improve overall safety and compliance.

Literature Review

This project is based on research in Natural Language Processing and Transformer models.

The first important paper is BERT, created by Devlin and others. BERT is a language model that reads text in both directions (left to right and right to left). This helps the model understand the meaning of words in context. BERT improved many NLP tasks, especially text classification.

The second important model is DistilBERT, created by Sanh and others. DistilBERT is a smaller and faster version of BERT. It uses fewer resources but keeps strong performance. This makes it useful for projects that run on limited hardware.

The third area of research is NLP in compliance and risk detection systems. Many studies show that NLP can be used to automatically analyze documents and detect errors or violations. This is useful in industries like finance, healthcare, and transportation.

These research works helped guide this project. BERT and DistilBERT influenced the choice of model, and NLP compliance research helped define the problem and approach.

Dataset

The dataset used in this project contains text examples of trucking log entries. Each example is labeled with one of several categories.

The main labels are:

- SAFE
- HOS_VIOLATION
- FORM_ERROR
- MISSING_DATA
- INSPECTION_ISSUE

The dataset is based on FMCSA rules. It includes different types of log entries to help the model learn how to classify them correctly.

Methodology

This project uses a supervised learning approach.

First, the data is cleaned and prepared. Then the text is converted into tokens using a tokenizer. Tokenization transforms text into numbers so that the model can process it.

After that, the dataset is split into training and testing sets. The model is trained using the training data and then evaluated using the test data.

The model used in this project is DistilBERT, a Transformer-based model. It learns patterns in text using attention mechanisms and predicts the correct class for each input.

Model

The model used is DistilBERT.

DistilBERT is a smaller and faster version of BERT. It is efficient and works well for classification tasks. The model reads the input text and assigns one of the predefined labels.

This model is suitable for this project because it balances performance and computational efficiency.

Results

The model achieved strong results on the test dataset.

- Accuracy: 94%
- Weighted F1-score: 0.94

The results show that the model performs very well overall.

The model is especially good at detecting important violations such as HOS violations and inspection issues. It also performs well on SAFE logs.

Some confusion was observed between similar classes like FORM_ERROR and MISSING_DATA. This is expected because these categories are similar and sometimes difficult to separate.

The model shows strong performance and can be used as a reliable tool for classification.

Why AUC-ROC wasn't used

AUC-ROC is a common evaluation metric, but it was not used as the main metric in this project. This is because the project focuses on a safety problem where missing a violation is very important.

In trucking compliance, it is more important to detect all violations (high recall) than to only have high overall accuracy. AUC-ROC can sometimes give high scores even if the model does not detect all violations well.

Instead, this project uses F1-score, precision, and recall. These metrics give a better understanding of how the model performs in real-world situations. The F1-score is especially useful because it balances precision and recall.

This choice makes the evaluation more realistic for a safety-critical application like trucking compliance.

Business Application

This system can be used in trucking companies to improve safety and efficiency.

The model can automatically check driver logs and detect violations. This reduces the need for manual review and saves time.

It can also help companies:

- Reduce audit time
- Detect violations quickly
- Avoid FMCSA fines
- Improve safety scores

This system supports safety managers by providing quick and accurate analysis of log data. It does not replace human workers but helps them work more efficiently.

Limitations

There are some limitations in this project:

First, the dataset is small. This can affect the performance of the model. Second, the data is synthetic, which means it is not real-world data. Because of this, the model may perform differently on real data.

Another limitation is that the model may confuse similar classes such as FORM_ERROR and MISSING_DATA.

Future Work

In the future, this project can be improved in several ways.

First, a larger and more realistic dataset can be used. Second, the model can be fine-tuned with real ELD data. Third, the accuracy of the model can be improved with better training techniques.

Another improvement is to build a web application so that users can easily upload and check log entries.

Conclusion

This project shows how NLP and Transformer models can be used to solve real-world problems.

The DistilBERT model successfully classifies trucking log entries and detects violations with high accuracy. The system can help trucking companies improve safety, reduce manual work, and increase efficiency.

This project demonstrates the practical use of AI in the trucking industry and shows how machine learning can support business operations.

References

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